

REROBOT

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Summary

The REROBOT program establishes a Japan-based rare earth recycling ecosystem leveraging the nation's vast stock of industrial robots. The core approach is a breakthrough in "disassembly tech": mechanizing and automating the motor-to-magnet extraction process—currently the most critical bottleneck—to improve efficiency by more than 10-fold. Over four years, six to ten global teams will utilize a 2 billion JPY budget. This initiative will ultimately launch a global spin-out startup, aggregating international feedstocks to secure critical material sovereignty.

The Challenge

As global decarbonization and electrification accelerate, green industries increasingly rely on **rare earth elements (REEs)** as foundational raw materials. However, highly monopolized global REE supply chains pose severe economic and geopolitical security risks. Japan's green industrial base is exceptionally vulnerable to these supply disruptions. According to de Cunzio et al. (2025), Japan's exposure to REE supply risks deteriorated drastically from the 11th least vulnerable nation in 2000 to the 3rd most vulnerable globally by 2015 [1]. Mitigating this systemic risk demands a multi-pronged strategy integrating resource conservation, alternative material development, and robust recycling. Yet, industrial-scale REE recycling remains commercially unviable worldwide.

While multiple barriers hinder effective REE recovery from end-of-life (EOL) products [2], the two most critical bottlenecks are: (1) the chronic shortage and supply instability of EOL feedstocks available in large, consistent volumes, and (2) the prohibitive labor costs and technical difficulties associated with disassembling high-power electric motors.

The Breakthrough

While electric vehicles (EVs), wind turbines, and hard disk drives are widely recognized for their use of rare earth magnets, industrial electric motors represent an equally vast, if not larger, unexploited reservoir of these critical materials. A prime example lies in industrial robotics. Japan ranks second globally in both the annual installation and the cumulative operational stock of industrial robots, having sustained the world's leading stock until 2015 [3,4].

The **REROBOT** program aims to establish a robust, Japan-based ecosystem to enable commercially viable REE recycling and permanent magnet remanufacturing. The primary strategic innovation of this program is the mechanization and automation of industrial motor disassembly.

As a key strategy to realize this innovation, the program leverages Japan's strategic advantage in its large stock of motors. Rather than relying on fragmented sourcing, the program will centralize a large volume of EOL motors—primarily from Japanese industrial robots—at a dedicated hub. This strategic aggregation directly solves the foundational market failure by achieving the economies of scale required for both rigorous technical demonstration and future commercial viability.

The program will then convene multiple collaborative teams specializing in database development and equipment engineering, with the goal of fostering breakthroughs that accelerate motor disassembly and magnet extraction by more than 10-fold compared to state-of-the-art technologies. The program will focus on the following two key technological developments:

1. Constructing the Robot Database: The participants will construct a comprehensive inventory detailing motor architectures and the raw material composition profiles of industrial robots and associated machinery. These collected and cataloged EOL assets will serve as a shared resource, fueling collaborative R&D across all participating program teams.

2. Disassembly Tech Innovation: REROBOT directly tackles the most prohibitive barrier to commercialization: the high cost of motor disassembly. While conventional REE recycling research heavily focuses on downstream chemical extraction from already-isolated magnets [2,5], the upstream labor-intensive process of extracting magnets from enclosed motor assemblies remains entirely unaddressed. For instance, removing rare earth magnets from a single EV motor currently demands up to three hours of manual, expert labor [6]. By advancing high-throughput motor disassembly automation, this program will transform REE recycling from a costly manual chore into a highly scalable, profitable industrial process.

The “Big if True” Impact

If successful, the REROBOT program will fundamentally redefine the economics of permanent magnet sustainability. Rather than focusing on downstream chemical extraction, the program’s unique value lies in its concentrated investment in the automated motor disassembly process. This program targets both the direct remanufacturing route (recycling EOL magnets as magnets) and the resource reclamation route (recovering elements from them).

A wide range of automated disassembly approaches are envisioned. One prime example is developing an automated line that combines low-cost demagnetization technology to neutralize powerful magnetic forces, followed by mechanical disassembly. Furthermore, we welcome the application and integration of diverse technological pathways for this disassembly stage, including hydrometallurgical, pyrometallurgical, hydrogen processing, and electric pulse methodologies. While we remain open to low-energy approaches that bypass disassembly altogether, we also anticipate technological innovations to scale up pyrometallurgical processes, including induction melting.

The ultimate goal of this program is to mechanize, automate, and scale this entire process up to an industrial level, utilizing energy efficiency as a secondary, complementary metric for technical evaluation. Successfully realizing this *low-energy automated motor disassembly*, combined with the establishment of large-scale motor collection logistics, will position Japan as a global hub for magnet recycling. Achieving these goals will dramatically mitigate the geopolitical risks associated with the international trade of REEs, while significantly advancing the resilience and competitiveness of Japan’s green industries.

The Plan

A Venture Director (VD) will oversee the strategic direction of the entire initiative. The “performers”—including startups, academic and research laboratories, and corporate new business development teams—will be openly recruited from all over the world. To ensure a closed-loop ecosystem, the program will actively engage and collaborate with Japanese industrial robot manufacturers, their end-users, and rare earth magnet refiners. From these applicants and partners, six to ten high-potential teams—including domestic Japanese teams, international teams, and joint/collaborative teams combining Japanese and overseas talent—will be selected to receive funding. A total budget of 2 billion JPY (approximately 13 million USD) will be allocated across the VD’s management and these selected performers. The structured four-year execution plan is detailed in the table.

Upon completion of this four-year program, the scale-up of the selected startups will be accelerated, or a new spin-out startup will be launched, leveraging the specialized talent, technological breakthroughs, and collective expertise accumulated throughout the initiative. Today, global startups are already revolutionizing the recycling and circular economy sectors; notable examples include Redwood Materials, which scales EV battery collection to recycle lithium, and Neste, which aggregates worldwide waste oil to produce Sustainable Aviation Fuel (SAF). As a definitive

outcome of this program, the foundation for an international REE recycling startup—poised to stand shoulder to shoulder with these global giants—will be established within four years.

Phase	Duration	Key Activities	Milestones
1 <i>Hub Development</i>	<i>Years 0-1</i>	Establish a domestic recycling yard. Collect decommissioned industrial robots. Disassemble various models to analyze and document physical structures and REE contents.	20+ tons of end-of-life robots collected. A comprehensive structural and REE-content database established for major industrial robot models.
2 <i>Technology Validation</i>	<i>Years 1-3</i>	Develop a low-cost demagnetization and mechanical separation technology. Validate diverse recycling pathways, including pyrometallurgical and hydrogen processing.	An automated disassembly prototype line operational. Disassembly time reduced from 3 hours to 18 minutes. High-purity REEs or magnets successfully recovered.
3 <i>Startup Spin-out</i>	<i>Years 4+</i>	Design a global feedstock network. Optimize the automated line for high-throughput commercial operations. Establish commercial business models for corporate spin-out.	Develop a finalized commercial roadmap. Launch of a global recycling startup.

Electronic waste—such as used home appliances, PCs, and discarded EVs from developed countries—is often shipped to less developed countries. In these regions, labor-intensive processing methods harm the environment and human health, while REEs are scattered and lost within wastewater or impurities. By changing policies in developed nations, however, we can redirect this flow into a domestic or global closed-loop supply chain [7]. This program will establish a new ecosystem in Asia, addressing environmental pollution in developing countries, enabling the efficient utilization of natural resources, and ultimately contributing to Japan’s economic security.

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